HIGGS BOSON DETECTION

Objective:

The objective of this project is to develop an effective machine learning model to classify particle collision events recorded by the ATLAS detector at CERN as either **signal events**—indicating occurrences of the Higgs boson—or **background events**, which represent other common particle interactions. The precise detection of the Higgs boson is fundamental in particle physics as it confirms the mechanism through which particles acquire mass, a keystone of the Standard Model(The **Standard Model** is a fundamental theory in particle physics that describes three of the four known fundamental forces in the universe: **electromagnetic, weak, and strong interactions**, but it excludes gravity.).

Given the complexity and high dimensionality of collision data, and the rarity of true Higgs boson events amid overwhelming background noise, this project leverages advanced machine learning techniques to improve classification accuracy. The goal is to minimize missed detections (false negatives) of the Higgs boson while maintaining low false positives, facilitating reliable discovery from experimental data

Basic physics to know about this project:

The Higgs boson is a fundamental particle predicted by the Standard Model and experimentally confirmed in 2012 by the ATLAS and CMS collaborations at the Large Hadron Collider (LHC). Particle collisions within the LHC generate numerous complex events, each producing numerous measurable signatures such as energies, momenta, and angular distributions of decay products.
Key physics concepts relevant here:
Signal vs. Background:
Signal: Events consistent with Higgs boson decay signatures, such as Higgs decaying into tau leptons.
Background: Standard particle processes that mimic these signatures but do not involve the Higgs boson.
Feature Representation:
High-level derived physics features (e.g., invariant mass, transverse momentum, angular

separations) summarize interactions crucial for differentiating signals.
Challenge of Rare Event Detection:
Signal events are far fewer than background, making classification akin to a rare-event detection problem requiring models with strong sensitivity (recall).
Importance of Accurate Classification:
Identifying Higgs bosons amidst noise validates physical theories and guides further experimental research—errors lead to missed discoveries or false claims.
This problem offers a fertile ground for ML, which can learn subtle complex patterns from multidimensional physics variables, helping physicists sift through huge datasets efficiently.

Experiments attempted:

Models Experimented

- **Random Forest Classifier:**
  Used as a baseline ensemble method, it yielded robust classification with around 84% accuracy and competitive precision and recall.
- **Neural Network (Deep Learning):**
  A fully connected Keras model was trained with dropout and early stopping, achieving respectable performance (~82% accuracy) and strong ROC AUC, but lagging behind tree-based models on recall for signal events.
- **XGBoost Classifier:**
  Gradient boosting-based model that matched or slightly exceeded Random Forest accuracy (84%) and outperformed both in recall (0.74 for signal), crucial for reducing missed Higgs detections.

Why Finalize on XGBoost?

- **Superior Recall for Signal Events:** Essential for Higgs detection where missing a true event is costly.
- **Balanced Overall Accuracy and F1-Score:** No compromise on classification quality.
- **Robustness and Interpretability:** Handles feature interactions well and provides feature importance insights.
- **Efficiency:** Reasonable training time and scalability.

Given these advantages and the experimental outcomes, XGBoost was finalized as the primary model for the project.

Model training pipeline:

Data Preparation

- **Dataset:** ATLAS Higgs Boson Challenge data from CERN, comprising hundreds of thousands of simulated collision events with over 30 physics-informed features.

- **Preprocessing:**
  - Dropped non-predictive columns (EventId, weights, etc.).
  - Checked for missing values (none found; known -999 sentinels appropriately handled).
  - Scaled features using StandardScaler to normalize magnitude disparities.
  - Mapped labels: ‘s’ → 1 (signal), ‘b’ → 0 (background).
  - Data split into 70% train / 30% test sets, stratified to preserve class distribution.

XGBoost Model Training

- **Parameters:**
  - Objective: binary logistic regression
  - Evaluation metric: logloss
  - Random state set for reproducibility
  - Default or lightly tuned other hyperparameters.
- **Training:**
  Model trained on the scaled training data until convergence.
- **Evaluation:**
  - Predictions made on held-out test data.
  - Performance assessed via accuracy (0.84), precision, recall, F1-score, and confusion matrix.
  - Attention focused on recall for class 1 (signal), achieving 0.74.
  - Feature importance analysis to identify top contributing physics features.

Future improvements:

- **Hyperparameter Tuning:** Systematic grid or Bayesian optimization over learning rate, tree depth, number of estimators, subsampling rates, and regularization could boost model robustness and recall further.
- **Advanced Feature Engineering:** Incorporate domain-driven feature transformations or interactions beyond the provided features to highlight subtle signal characteristics.
- **Ensemble Methods:** Combine XGBoost with neural networks or other classifiers (stacking/blending) to exploit complementary strengths.
- **Deep Learning Enhancements:** Explore more complex architectures (e.g., convolutional or recurrent layers if representing sequential or spatial data), better class imbalance handling (weighted loss, focal loss), or data augmentation strategies.
- **Calibration and Uncertainty Estimation:** Incorporate probabilistic calibration to better quantify prediction confidence—valuable in experimental physics decisions.
- **Real Detector Data and Transfer Learning:** Extend models using real experimental data or adapt pretrained models from related physics tasks to improve generalization.

This final report captures the essence, methodology, results, and pathway forward for leveraging machine learning, focusing on XGBoost, to enable effective Higgs boson detection within a high-stakes, high-dimensional physical classification problem.